# ALCHEMY AOE ALLIANCE

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|---|---|---|
| **TYPE:**<br>COMPONENT SPECIFICATION FOR RANDOM MAP<br>**(CSRM)** | | **PROJECT:**<br>SUPER RANDOM<br>**(RAND)** |
| **CSRM-RAND-02** | | **REVISION: X1** |

## Pocket Random



## Table of Contents

# 1 - Referenced Documents

Table 1 below is a list of all specifications that are components to this one. Each of its entries were designed as an interchangeable part, to be used in context for this specification, or others like it:

| Specification Number | Nomenclature |
|---|---|
| GSRM-RAND-1 | General Requirements, Super-Randoms |

*Table 1: Bill of Materials*

If any part of any documents listed in Table 1 conflict with this specification, then this specification shall assume priority.

# 2 - Definitions

The following definitions apply:

- *Origin of Player Lands* – Location of town center.

# 3 - Overview

This component specification defines requirements for a Super-Random style map with random boundary to be used in an Alchemy AOE Competition.

# 4 - General Characteristics

The requirements of GSRM-RAND-1 apply unless superseded by this specification.

# 5 - Rule Changes

This section is intentionally left blank.

# 6 - Land Configuration

Pocket Random shall be arranged generally per Figure 6 below:

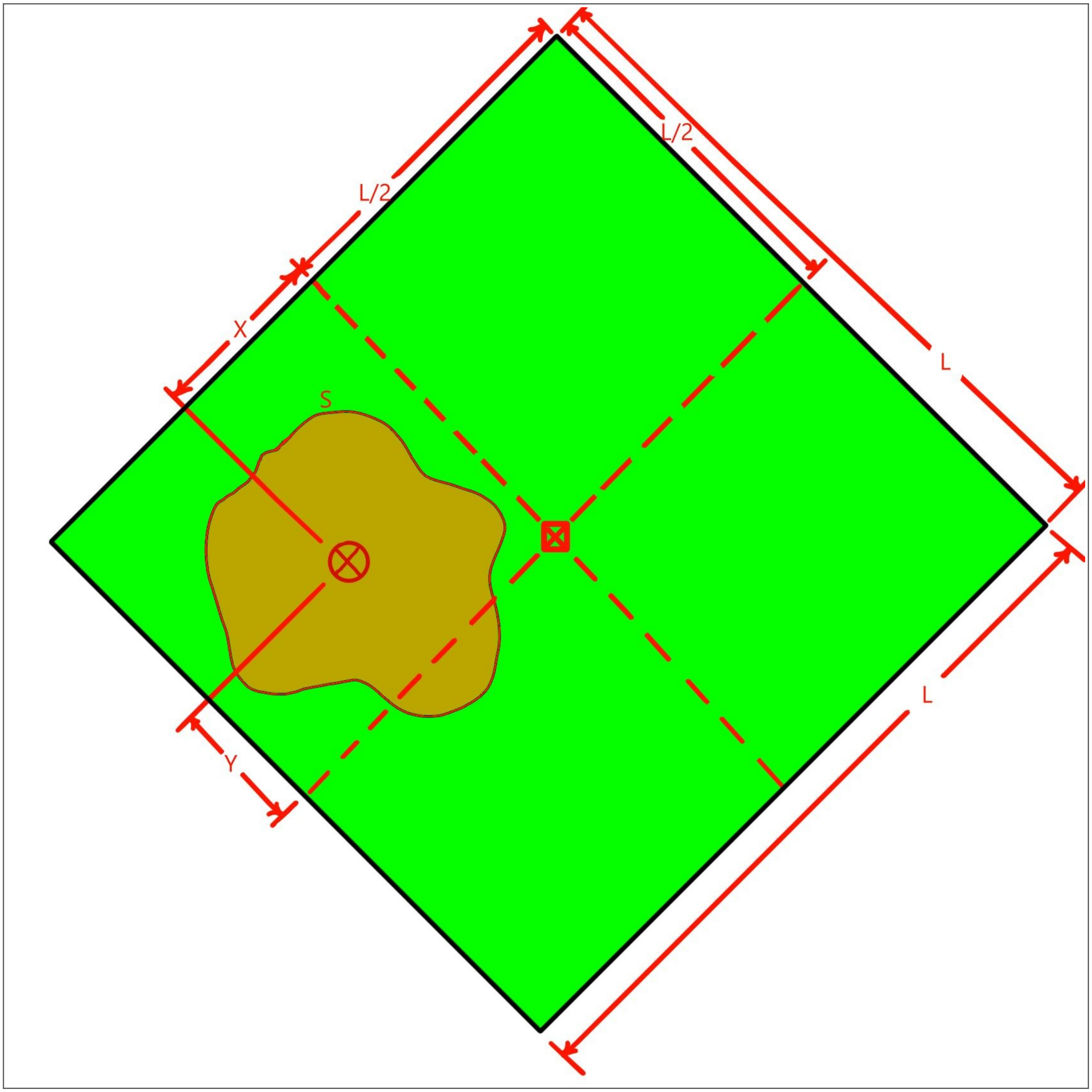


*Figure 6: Pocket Random General Configuration*

# 6.A - Zones

In this way, two zones are established, described as follows:

## 6.A.1 - Pocket

Brown area within red arbitrary boundary “*S*”.

## 6.A.2 - Boundary

Red curve labeled “*S*”, containing the Pocket.

## 6.A.3 - Outland

Any area not belonging to the aforementioned zones.

# 6.B - Origin of Player Lands

The *Origin of Player Lands* for each player shall reside within the Pocket. The average Origin of Player Lands is indicated by the red circumscribed X, and shall not correspond with the geometric center of the Pocket in more than 10% of theoretical generations.

# 6.C - Domain Sizing

## 6.C.1 - Base Map

The random map shall disregard size selection in lobby and override depending on detected player quantity. Dimension “*L*” indicated in Figure 6 shall depend on the number of players:

| Player Count | Edge Length [tiles] | |
|---|---|---|
| | **Minimum** | **Maximum** |
| 1 and 2 | 192 | 240 |
| 3 and 4 | 269 | 336 |
| 5 and 6 | 320 | 400 |
| 7 and 8 | 384 | 480 |

*Table 6.C.1: Map Base Scaling*

Each integer value between minimum and maximum shall have equal probability of random selection.

### 6.C.2 - Pocket Area

The boundary of the Pocket shall encompass an approximate area between 20% and 45% of the domain from Table 6.C.1. No fewer than six sizes within this range shall be possible with approximately equal probability.

Absolute Pocket Area shall be a derivative of relative space and Base Map sizing variations, determined through tolerance stack-up. For example, on a map scaled for 1 and 2 players, the minimum Pocket area would be (0.20)*(192)^2 = 7373 sq tiles, whereas the maximum Pocket area would be (0.45)*(240)^2 = 25920 sq tiles. Continuing from this example, the probability of either size extreme would be [192, 193, … 240] – vector length 49 – multiplied by [20, 25, … 45] – vector length 6 – for a chance of one in 6*(49) = 294 or ~0.3%.

## 6.D - Pocket Randomizations

### 6.D.1 - Positioning

Per Figure 6 the *Origin of Player Lands* shall have cartesian offset from the center of the base map, indicated by [X] at [L/2, L/2]. Offsets "X" and "Y" shall not equal zero for more than 10% of theoretical generations, respectively. In this way, coincidence of the *Origin of Player Lands* and the map center shall occur in no more than 1% of generations.

### 6.D.2 - Shape

All shapes for the Pocket are allowed except those described in Figure 6.D.2.

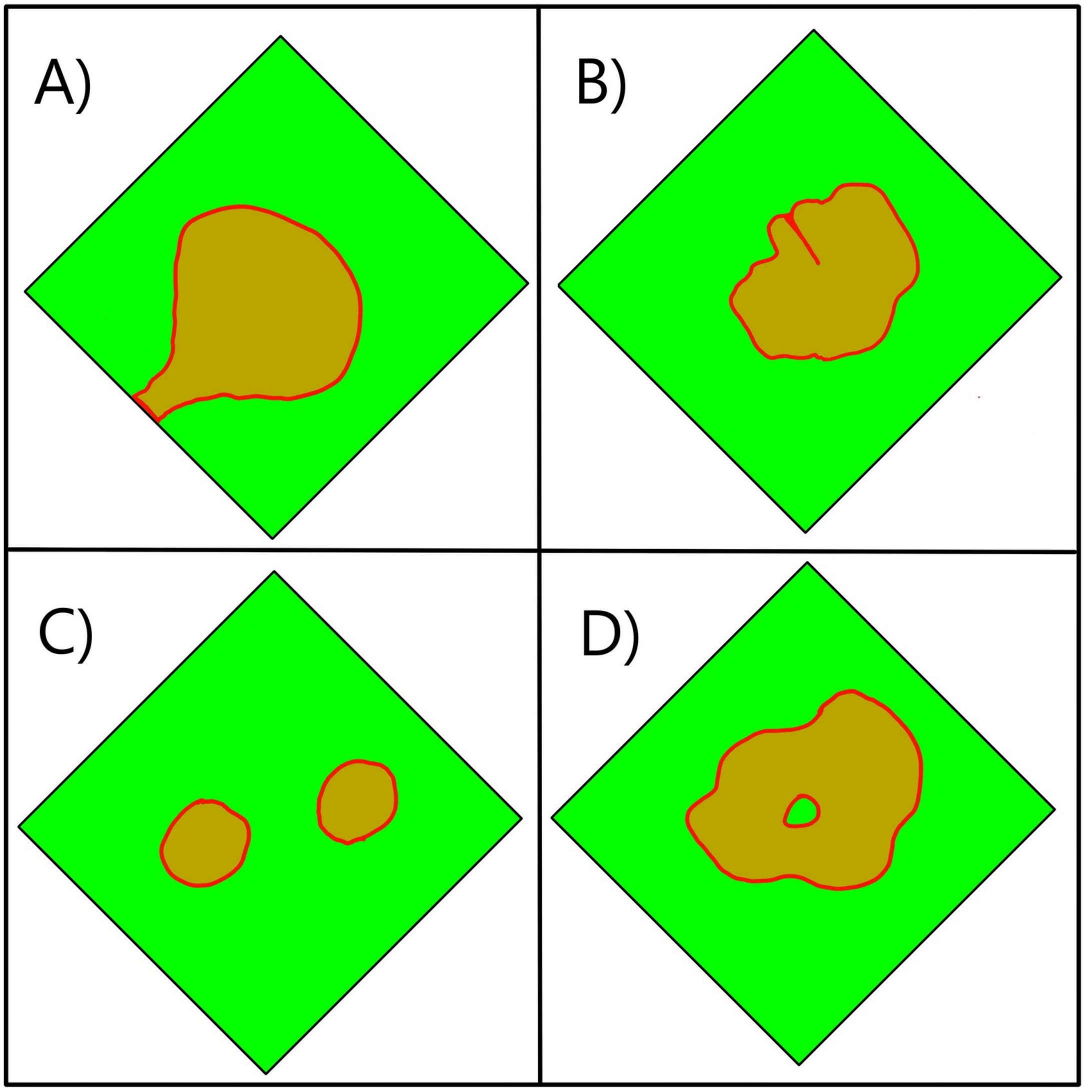


*Figure 6.D.2: Forbidden Pocket Configurations*

Forbidden configurations include:

A) Intersection of Pocket with edge of default game space.

B) Inclusion of Boundary into Pocket.

C) Isolated sections of Pocket.

D) Outland contained within Pocket.

The random map shall maximize Pocket shape variation within the above listed constraints.

# 7 - Characteristics by Zone

## 7.A - Pocket

Pocket terrain shall be open and unbuildable within 2 tiles from the boundary, but assume the elevations, terrains, objects, etc. of a Super Random style map per GSRM-RAND-1 everywhere else.

## 7.B - Boundary

The boundary shall reflect the elevations and terrains of the Pocket, but populated by blocking objects. These objects shall permanently prevent access to the Outland from the Pocket.

## 7.C - Outland

The Outland shall reflect the neutral spaces of the Pocket in elevation, terrain, and gaia objects. Since this area is inaccessible and non-interactive to players, it shall be designed with "performance" as the top priority. Example strategies include:

- All objects in this zone placed after objects in Pocket zone, including forest trees, thereby mitigating burden of objects in Pocket that "avoid_forest_zone X".
- Minimal wildlife (boar, deer, etc.) or wildlife placed by giving graphics to dummy objects that do not move.
- Minimal gaia buildings with moving animations, such as mills.
- Minimal use of actor areas, particularly those above radius 2.

The Outland may consist of uniform black terrain instead of thematic decoration to improve computer handling/performance if required.

# 8 - Variation Count

The script file shall be at least 2 MB in size when filled with pre-computed Pocket shapes. The shapes shall be of maximum efficiency to individually consume the least amount of space and therefore allow maximum total count of all shapes.

# Signatures:

TechChariot 2025-10-29

*Prepared* *Date (YYYY-MM-DD)*

SampleChecker 2025-MM-DD

*Checked* *Date (YYYY-MM-DD)*

SampleApprover 2025-MM-DD

*Approved* *Date (YYYY-MM-DD)*

# Revision History:

| Revision | Description | Change Document | Date (YYYY-MM-DD) |
|---|---|---|---|
| X1 | ------ Advanced Release for Procurement ------ | N/A | 2025-11-10 |